AIR CONDITIONING & REFRIGERATION (RAC) COMPREHENSIVE TECHNICAL HANDBOOK

Module: Split AC, Frost-Free Systems, Inverters & Refrigerant Management

UNIT 1: SPLIT AIR CONDITIONER SCHEMATICS & WIRING

TERMINAL DESIGNATIONS & SIGNAL CONNECTIONS:
1 (L) - Indoor Main Live Supply from Distribution Board via 16A/20A Type C MCB.
2 (N) - Neutral Common return line shared with indoor fan motor, transformer and outdoor fan.
3 (Comp / Signal) - Switched Live relay output from indoor PCB micro-controller (comp command).
Earth (PE) - Continuous equipment grounding connecting indoor chassis and outdoor condenser casing.
20-SECOND SAFETY TIME DELAY: Protects rotary compressor against high head pressure re-start.

UNIT 2: REFRIGERANT RECOVERY SYSTEM & PROCEDURES

EQUIPMENT SETUP & SAFE OPERATING PROTOCOL:
* Manifold Blue (Compound) Hose connects to Suction Service Port of condensing unit.
* Center Yellow Hose connects to Recovery Machine INLET.
* High Pressure Red Hose connects from Recovery Machine OUTLET to Vapor valve of Recovery Cylinder.
* Tare Weight Tare Verification: Measure cylinder on digital charging scale to prevent hydraulic over-fill.
* Self-Purge Procedure: Rotate machine valves to PURGE to draw remaining liquid out of condenser coil.

UNIT 3: REFRIGERATOR FROST-FREE DEFROST SYSTEM

4-PIN DEFROST TIMER INTERNAL CONFIGURATION:
Pin 1: Timer Motor Line 1 (Power Return via Defrost Heater in Cooling mode).
Pin 3: Timer Common Input from Temperature Thermostat.
Pin 4: Cooling Contact energizes Compressor Relay and Evaporator Fan Motor (Runs ~6 to 8 hours).
Pin 2: Defrost Contact energizes Glass Tube/Sheath Defrost Heater through Bi-metal (-5 deg C closing).
Thermal Fuse (72 deg C to 77 deg C): High safety limit fuse wired in series with heater against runaway fires.

TROUBLESHOOTING MATRIX FOR RAC TECHNICIANS

Common Fault 1: Indoor Blower Runs, Outdoor Compressor Does Not Start

- Check 230V at Terminal 3 (Signal) relative to Terminal 2 (Neutral). If 0V, PCB relay or sensor faulty.
- If 230V present, inspect outdoor dual run capacitor (35uF/450V) with capacitance meter.
- Measure compressor winding resistance (C-S + C-R = S-R). Check for continuity to ground.

Common Fault 2: Refrigerator Freezer Coils Frosted Solid, Lower Cab Warm

- Test Bi-metal thermostat continuity while frozen in ice bath (0 Ohms below -5 deg C).
- Test Defrost Heater element resistance (typically 150 to 300 Ohms; infinite indicates open circuit).
- Manually advance defrost timer cam with screwdriver; verify timer motor advances automatically.